

HDOC DAY-36 ASSIGNMENT

Menu-Driven String Operations Using Separate Functions

1. Problem Understanding

The task is to write a menu-driven C program that reads a string and performs the required operations using separate functions.

- Count the number of characters in the string without using a built-in length function.
- Print each character of the string on a new line.
- Count the vowels and consonants in the string.
- Continue displaying the menu until the user selects Exit.

2. Standard Test Cases

The test cases include all required inputs: menu choice and the complete input string.

TC	Choice	Input String	Expected Result
TC1	1	Hello	Number of characters = 5
TC2	1	Hello World	Number of characters = 11
TC3	2	Cat	C, a and t printed on separate lines
TC4	3	Education	Vowels = 5, Consonants = 4
TC5	3	C Programming	Vowels = 3, Consonants = 9
TC6	4	Hello	Program exits

3. Algorithm

A. Common Menu Logic

1. Start.
2. Read a string from the user.
3. Display the menu options.
4. Read the user choice.
5. Use switch-case and call the corresponding function.
6. Repeat the menu until the user selects Exit.
7. Stop.

B. Count Characters

1. Initialize count = 0.
2. Start from the first character of the string.
3. Continue until the null character is reached.
4. Increase count for every character.
5. Display count.

C. Print Each Character

1. Start from the first character.

2. Continue until the null character is reached.
3. Print the current character followed by a new line.

D. Count Vowels and Consonants

1. Initialize vowels = 0 and consonants = 0.
2. Check each character of the string.
3. If the character is an alphabet, check whether it is a vowel.
4. Increase the vowel count for a vowel; otherwise increase the consonant count.
5. Ignore spaces, digits and special characters.
6. Display both counts.

4. C Program

```
#include <stdio.h>
```

```
void countCharacters(char str[])
```

```
{
    int i = 0;
    while (str[i] != '\0')
    {
        i++;
    }
    printf("Number of characters = %d\n", i);
}
```

```
void printCharacters(char str[])
```

```
{
    int i = 0;
    while (str[i] != '\0')
    {
        printf("%c\n", str[i]);
        i++;
    }
}
```

```
void countVowelsConsonants(char str[])
```

```
{
    int i = 0, vowels = 0, consonants = 0;
    char ch;
    while (str[i] != '\0')
    {
```

```

    ch = str[i];
    if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z'))
    {
        if (ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U' ||
            ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
            vowels++;
        }
        else
        {
            consonants++;
        }
    }
    i++;
}

printf("Vowels = %d\n", vowels);
printf("Consonants = %d\n", consonants);
}

int main()
{
    char str[100];
    int choice;
    printf("Enter a string: ");
    scanf(" %99[^\n]", str);
    do
    {
        printf("\nMENU\n");
        printf("1. Count characters\n");
        printf("2. Print each character on a new line\n");
        printf("3. Count vowels and consonants\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:

```

```

        countCharacters(str);

        break;
    case 2:
        printCharacters(str);

        break;
    case 3:
        countVowelsConsonants(str);

        break;
    case 4:
        printf("Exiting program.\n");

        break;
    default:
        printf("Invalid choice.\n");

    }
}

while (choice != 4);

return 0;

}

```

5. Compilation and Execution Evidence

```

kali@kali: ~/Desktop
(kali@kali)-[~/Desktop]
$ ./question
Enter a string: Hello

MENU
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice:
1
Number of characters = 5

MENU
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 4
Exiting program.

(kali@kali)-[~/Desktop]
$ ./question
Enter a string: Hello World

MENU
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 1
Number of characters = 11

MENU
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 4
Exiting program.

```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ ./question
Enter a string: Cat

MENU
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 2
C
a
t

MENU
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 4
Exiting program.

(kali@kali)-[~/Desktop]
$ ./question
Enter a string: Education

MENU
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 3
Vowels = 5
Consonants = 4

MENU
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 4
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
3. Count vowels and consonants
4. Exit
Enter your choice: 4
Exiting program.

(kali@kali)-[~/Desktop]
$ ./question
Enter a string: C Programming

MENU
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 3
Vowels = 3
Consonants = 9

MENU
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 4
Exiting program.

(kali@kali)-[~/Desktop]
$ ./question
Enter a string: Hello

MENU
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 4
Exiting program.

(kali@kali)-[~/Desktop]
$
```

6. Verification of All Test Cases

TC	Choice	Input String	Actual Result	Status
TC1	1	Hello	Number of characters = 5	Pass
TC2	1	Hello World	Number of characters = 11	Pass
TC3	2	Cat	C, a and t on separate lines	Pass
TC4	3	Education	Vowels = 5, Consonants = 4	Pass

TC5	3	C Programming	Vowels = 3, Consonants = 9	Pass
TC6	4	Hello	Program exits	Pass